hundreds of partners and often breaks when roles, attributes, or organizational structures change.

Just as chapter 12 emphasized the importance of trustworthy context in authorization decisions, a similar need exists here. Context isn't limited to device posture, session details, or environmental factors. It also includes organizational roles, professional licenses, clinical affiliations, supplier relationships, and project-specific delegations. These are forms of context that ACME cannot generate or maintain on its own, yet they can be just as vital as device or session attributes when assessing an access request. Viewing verifiable credentials as another category of trusted signals highlights an important idea: external attestations can be integrated into the authorization engine using the same architectural model and policy-driven approach described earlier.

Verifiable credentials offer a structured way to gather external context in a trustworthy and verifiable format. Instead of relying on internal accounts or federated assertions, ACME's authorization system can assess credentials issued by organizations that possess the relevant facts. These credentials represent roles, qualifications, delegations, and relationships as portable, cryptographically verifiable statements. They naturally integrate into ACME's existing context model because they serve as another type of trustworthy signal that the authorization engine can use to make decisions.

The rest of this chapter explains how verifiable credentials are organized, how ACME can decide whether to trust an issuer, and how these credentials can provide reliable signals to support authorization decisions across the collaboration platform.

16.2 How verifiable credentials work

Verifiable credentials package identity attributes into cryptographically secured claims that people can carry on their own devices and present when required. Some people are surprised that this could work. We've been trained to believe that identity information must stay on a central server or flow directly between organizations. The idea that someone can carry authoritative attributes on their own device and reliably present them may